Predicting Booking Outcomes Feature Importance and Business Strategies for InnHotels

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About Data and its Dictionary

Dataset:

NNHotelsGroup

Data Overview:

The data contains the different attributes of customers' booking details.

Data Dictionary:

- Booking_ID: the unique identifier of each booking
- no_of_adults: Number of adults
- no_of_children: Number of Children
- no_of_weekend_nights: Number of weekend nights (Saturday or Sunday) the guest stayed or booked to stay at the hotel
- no_of_week_nights: Number of weeknights (Monday to Friday) the guest stayed or booked to stay at the hotel
- type_of_meal_plan: Type of meal plan booked by the customer:
    - Not Selected – No meal plan selected
    - Meal Plan 1 – Breakfast
    - Meal Plan 2 – Half board (breakfast and one other meal)
    - Meal Plan 3 – Full board (breakfast, lunch, and dinner)
- required_car_parking_space: Does the customer require a car parking space? (0 - No, 1- Yes)
- room_type_reserved: Type of room reserved by the customer. The values are ciphered (encoded) by INN Hotels Group
- lead_time: Number of days between the date of booking and the arrival date
- arrival_year: Year of arrival date
- arrival_month: Month of arrival date
- arrival_date: Date of the month
- market_segment_type: Market segment designation.
- repeated_guest: Is the customer a repeated guest? (0 - No, 1-

Yes)

- no_of_previous_cancellations: Number of previous bookings that were canceled by the customer prior to the current booking
- no_of_previous_bookings_not_canceled: Number of previous bookings not canceled by the customer prior to the current booking
- avg_price_per_room: Average price per day of the reservation; prices of the rooms are dynamic. (in euros)
- no_of_special_requests: Total number of special requests made by the customer (e.g. high floor, view from the room, etc)
- booking_status: Flag indicating if the booking was canceled or not.

Number of rows and columns:

```
The number of rows in data: 36275
The number of columns in data: 19
```

First Five Rows:

	Booking_ID	no_of_adults	no_of_children	no_of_weekend_nights	no_of_week_nights	type_of_meal_plan	required_car_parking_space	room_type_reserved	lead_time	a
0	INN00001	2	0	1	2	Meal Plan 1	0	Room_Type 1	224	
1	INN00002	2	0	2	3	Not Selected	0	Room_Type 1	5	
2	INN00003	1	0	2	1	Meal Plan 1	0	Room_Type 1	1	
3	INN00004	2	0	0	2	Meal Plan 1	0	Room_Type 1	211	
4	INN00005	2	0	1	1	Not Selected	0	Room_Type 1	48	

e	market_segment_type	repeated_guest	no_of_previous_cancellations	no_of_previous_bookings_not_canceled	avg_price_per_room	no_of_special_requests	booking_status
2	Offline	0	0	0	65.00	0	Not_Canceled
5	Online	0	0	0	106.68	1	Not_Canceled
3	Online	0	0	0	60.00	0	Canceled
0	Online	0	0	0	100.00	0	Canceled
1	Online	0	0	0	94.50	0	Canceled

DataFrame Information:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 36275 entries, 0 to 36274
Data columns (total 19 columns):
 #   Column                                Non-Null Count  Dtype
---  ------                                --------------  -----
 0   Booking_ID                            36275 non-null  object
 1   no_of_adults                          36275 non-null  int64
 2   no_of_children                        36275 non-null  int64
 3   no_of_weekend_nights                  36275 non-null  int64
 4   no_of_week_nights                     36275 non-null  int64
 5   type_of_meal_plan                     36275 non-null  object
 6   required_car_parking_space            36275 non-null  int64
 7   room_type_reserved                    36275 non-null  object
 8   lead_time                             36275 non-null  int64
 9   arrival_year                          36275 non-null  int64
 10  arrival_month                         36275 non-null  int64
 11  arrival_date                          36275 non-null  int64
 12  market_segment_type                   36275 non-null  object
 13  repeated_guest                        36275 non-null  int64
 14  no_of_previous_cancellations          36275 non-null  int64
 15  no_of_previous_bookings_not_canceled  36275 non-null  int64
 16  avg_price_per_room                    36275 non-null  float64
 17  no_of_special_requests                36275 non-null  int64
 18  booking_status                        36275 non-null  object
dtypes: float64(1), int64(13), object(5)
memory usage: 5.3+ MB
```

Checking Duplicate values:

```
0
```

Booking ID column:

```
1        INN00002
2        INN00003
3        INN00004
4        INN00005
           ...
36270    INN36271
36271    INN36272
36272    INN36273
36273    INN36274
36274    INN36275
Name: Booking_ID, Length: 36275, dtype: object>
```

Dropping Booking ID column:

	no_of_adults	no_of_children	no_of_weekend_nights	no_of_week_nights	type_of_meal_plan	required_car_parking_space	room_type_reserved	lead_time	arrival_year
0	2	0	1	2	Meal Plan 1	0	Room_Type 1	224	2017
1	2	0	2	3	Not Selected	0	Room_Type 1	5	2018
2	1	0	2	1	Meal Plan 1	0	Room_Type 1	1	2018
3	2	0	0	2	Meal Plan 1	0	Room_Type 1	211	2018
4	2	0	1	1	Not Selected	0	Room_Type 1	48	2018

Checking Null values:

```
Booking_ID                              0
no_of_adults                            0
no_of_children                          0
no_of_weekend_nights                    0
no_of_week_nights                       0
type_of_meal_plan                       0
required_car_parking_space              0
room_type_reserved                      0
lead_time                               0
arrival_year                            0
arrival_month                           0
arrival_date                            0
market_segment_type                     0
repeated_guest                          0
no_of_previous_cancellations            0
no_of_previous_bookings_not_canceled    0
avg_price_per_room                      0
no_of_special_requests                  0
booking_status                          0
dtype: int64
```

Target Column:

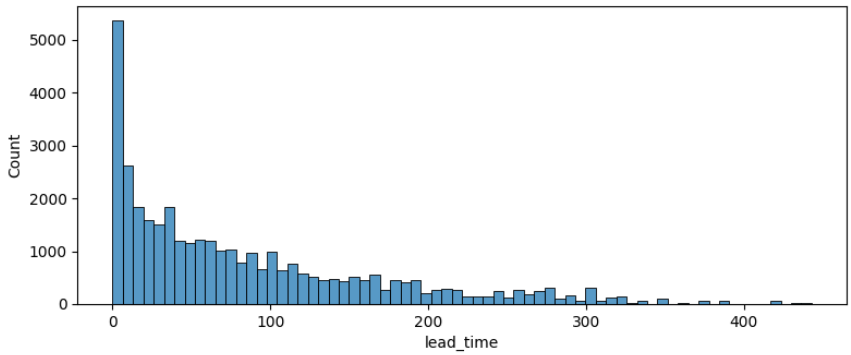
```
booking_status
Not_Canceled    24390
Canceled        11885
Name: count, dtype: int64
```

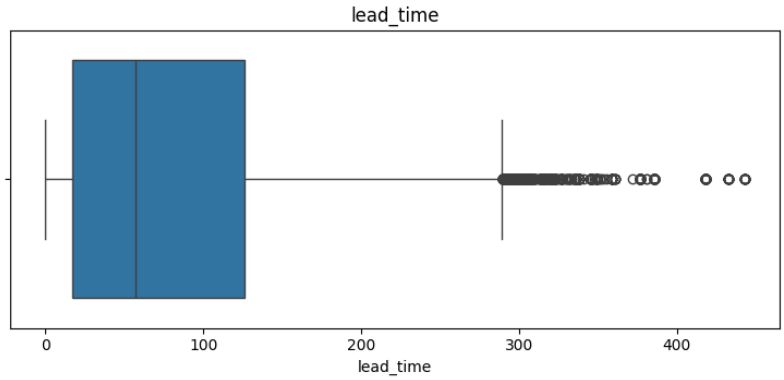
Statistical Information:

	count	mean	std	min	25%	50%	75%	max
no_of_adults	36275.0	1.844962	0.518715	0.0	2.0	2.00	2.0	4.0
no_of_children	36275.0	0.105279	0.402648	0.0	0.0	0.00	0.0	10.0
no_of_weekend_nights	36275.0	0.810724	0.870644	0.0	0.0	1.00	2.0	7.0
no_of_week_nights	36275.0	2.204300	1.410905	0.0	1.0	2.00	3.0	17.0
required_car_parking_space	36275.0	0.030986	0.173281	0.0	0.0	0.00	0.0	1.0
lead_time	36275.0	85.232557	85.930817	0.0	17.0	57.00	126.0	443.0
arrival_year	36275.0	2017.820427	0.383836	2017.0	2018.0	2018.00	2018.0	2018.0
arrival_month	36275.0	7.423653	3.069894	1.0	5.0	8.00	10.0	12.0
arrival_date	36275.0	15.596995	8.740447	1.0	8.0	16.00	23.0	31.0
repeated_guest	36275.0	0.025637	0.158053	0.0	0.0	0.00	0.0	1.0
no_of_previous_cancellations	36275.0	0.023349	0.368331	0.0	0.0	0.00	0.0	13.0
no_of_previous_bookings_not_canceled	36275.0	0.153411	1.754171	0.0	0.0	0.00	0.0	58.0
avg_price_per_room	36275.0	103.423539	35.089424	0.0	80.3	99.45	120.0	540.0
no_of_special_requests	36275.0	0.619655	0.786236	0.0	0.0	0.00	1.0	5.0

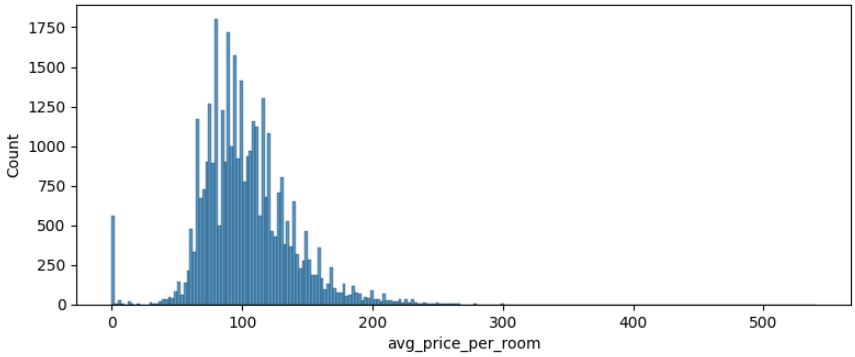
Exploratory Data Analysis

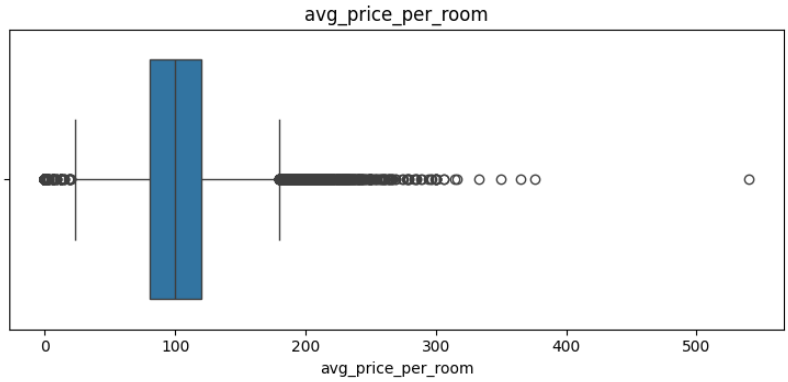
No. of Days b/w the date of booking and the arrival date (Lead time)





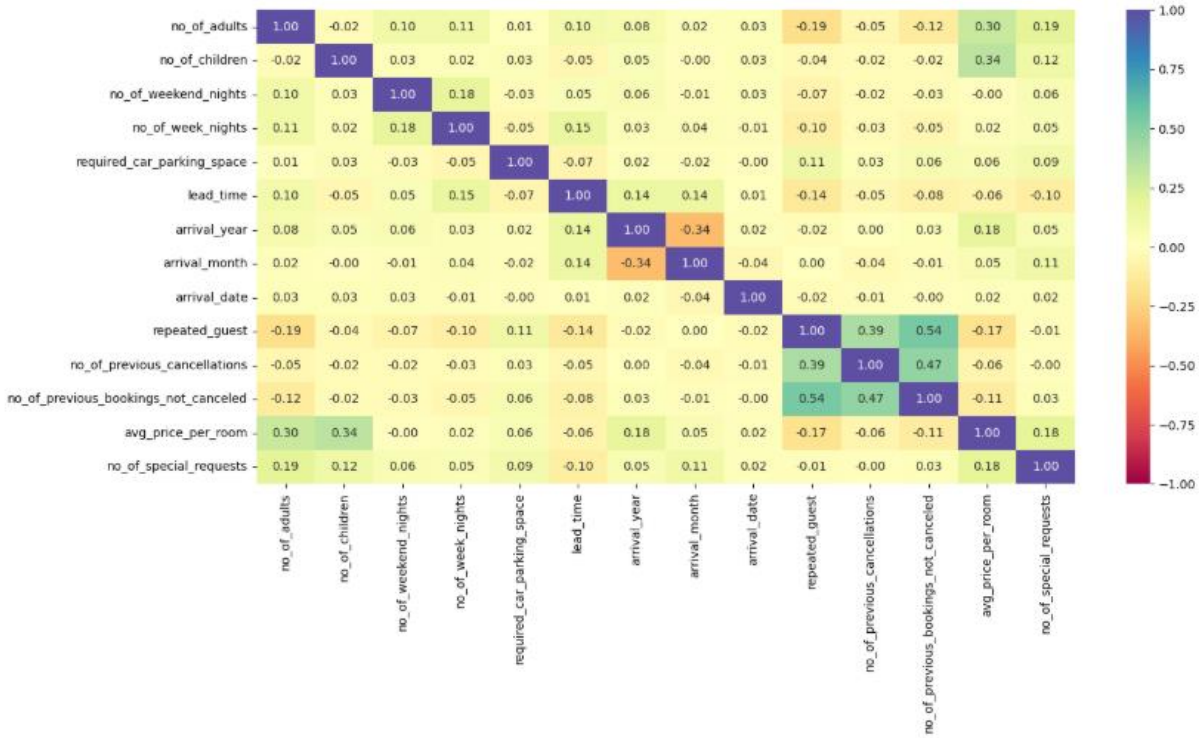
Average price per day of the reservation



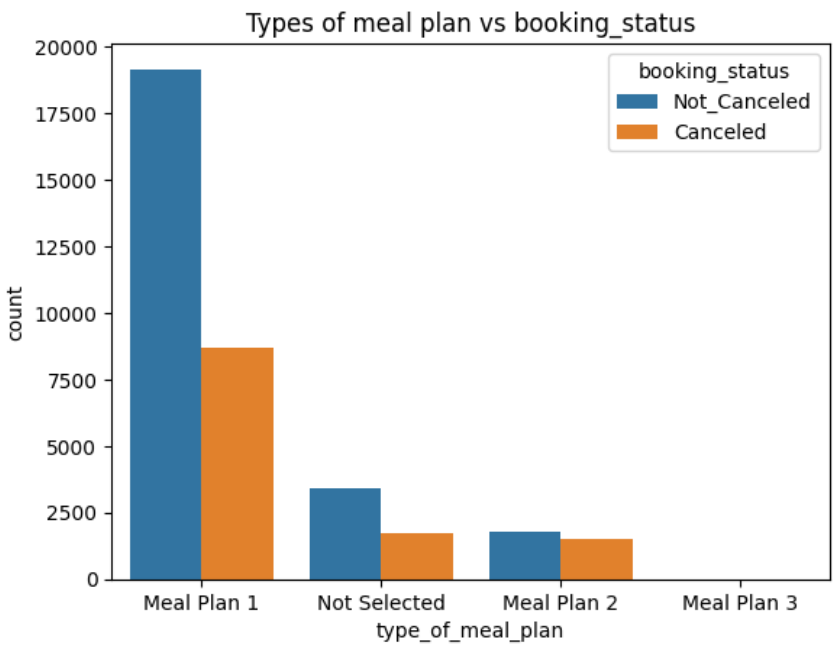


Booking Status

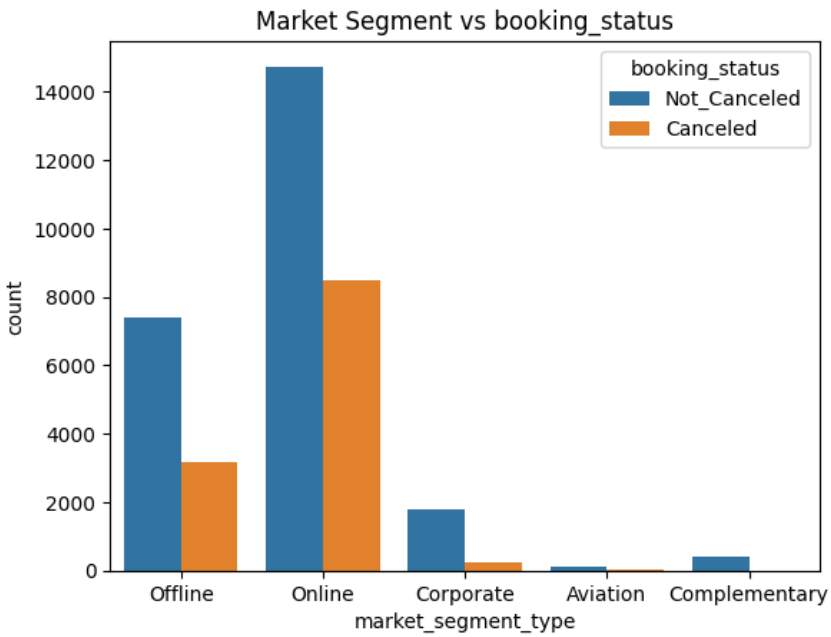
Bivariate Analysis



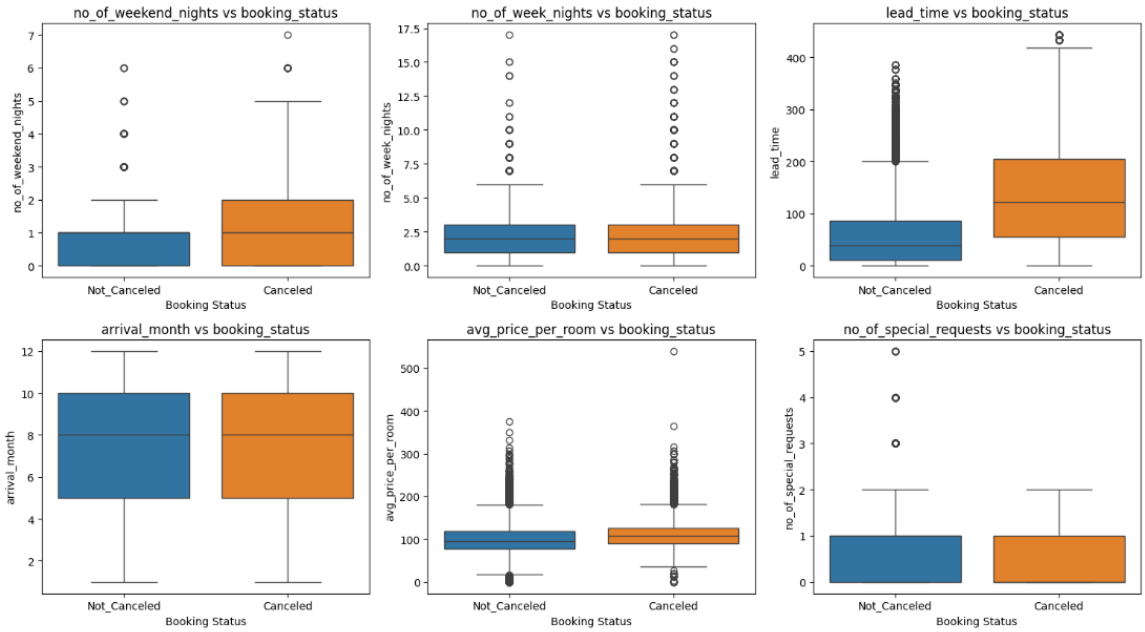
Types of meal plan vs booking status



Market Segment vs booking status



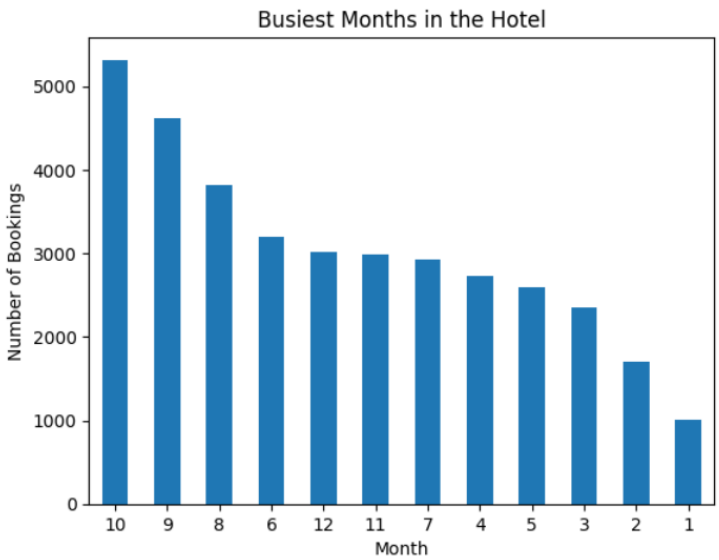
Booking Status vs Numerical Columns



EDA Questions and Answers

1. What are the busiest months in the hotel?

```
arrival_month
10    5317
9     4611
8     3813
6     3203
12    3021
11    2980
7     2920
4     2736
5     2598
3     2358
2     1704
1     1014
Name: count, dtype: int64
```



Observation:

The Busiest months in the hotel are as follows:

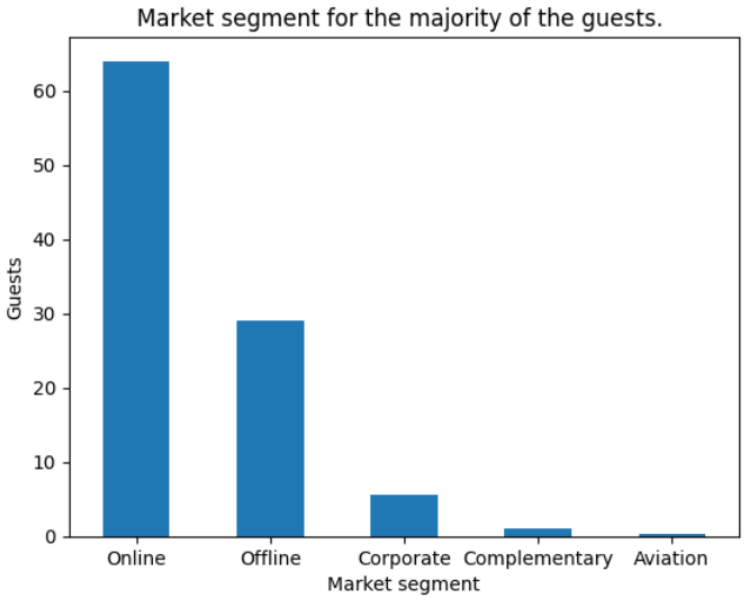
- October – 5,317 bookings
- September – 4,611 bookings
- August – 3,813 bookings

Insight:

The number of bookings for each month, with October being the busiest month followed by September and August.

2. Which market segment do most of the guests come from?

```
market_segment_type
Online           63.994487
Offline          29.022743
Corporate         5.560303
Complementary     1.077877
Aviation          0.344590
Name: proportion, dtype: float64
```



Observation:

The Market segment for the majority of the guests is the **Online** segment, with **64%**.

Insight:

Online segments play a crucial role in driving bookings for INN Hotels Group.

3. Hotel rates are dynamic and change according to demand and customer demographics. What are the differences in room prices in different market segments?

- Online customers have the highest mean price (112.26).
- Complementary bookings have a mean close to zero.
- Offline and Corporate segments also show slightly lower mean room prices compared to Online.
- Aviation customers have very stable prices (low standard deviation).

market_segment_type	count	mean	std	min	25%	50%	75%	max
Aviation	125.0	100.704000	8.538361	79.0	95.0	95.0	110.00	110.0
Complementary	391.0	3.141765	15.512972	0.0	0.0	0.0	0.00	170.0
Corporate	2017.0	82.911740	23.690001	31.0	65.0	79.0	95.00	220.0
Offline	10528.0	91.632679	24.995597	12.0	75.0	90.0	109.00	540.0
Online	23214.0	112.256855	35.220319	0.0	89.0	107.1	131.75	375.5

4. What percentage of bookings are canceled?

```
booking_status
Not_Canceled    67.236389
Canceled        32.763611
Name: proportion, dtype: float64
```

The percentage of bookings that are **Canceled** is **32.76%**.

5. Repeating guests are the guests who stay in the hotel often and are important to brand equity. What percentage of repeating guests cancel?

```
repeated_guest
0    99.865377
1     0.134623
Name: proportion, dtype: float64
```

Observation:

- Only 0.13% of cancellations are by repeating guests
- 99.87% of cancellations are by non-repeating guests

6. Many guests have special requirements when booking a hotel room. Do these requirements affect booking cancellation?

Observation:

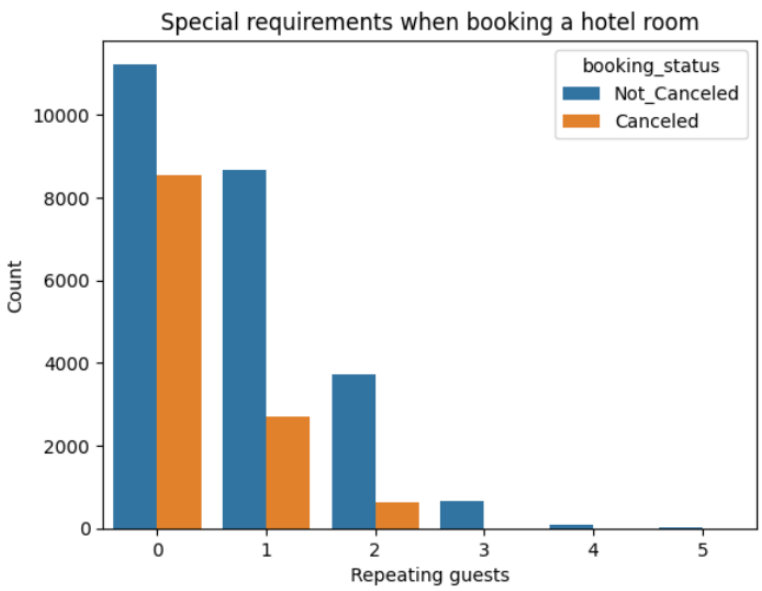
Guests who make special requests:

- Only 24% for 1 request.
- Only 15% for 2 requests.
- 0% for 3 or more.

Insight:

Guests with no special requests cancel more often — around 43% of these bookings are canceled. As the number of special requests increases, the cancellation rate drops significantly.

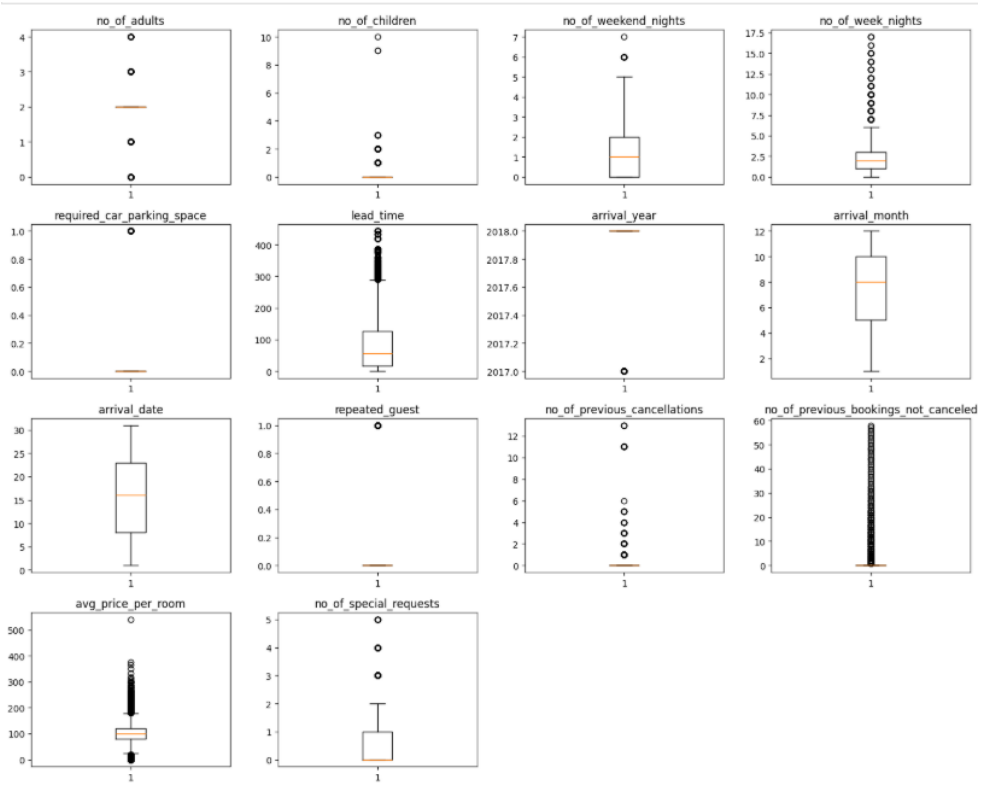
no_of_special_requests	booking_status	proportion
0	Not_Canceled	56.793245
	Canceled	43.206755
1	Not_Canceled	76.233184
	Canceled	23.766816
2	Not_Canceled	85.403300
	Canceled	14.596700
3	Not_Canceled	100.000000
4	Not_Canceled	100.000000
5	Not_Canceled	100.000000



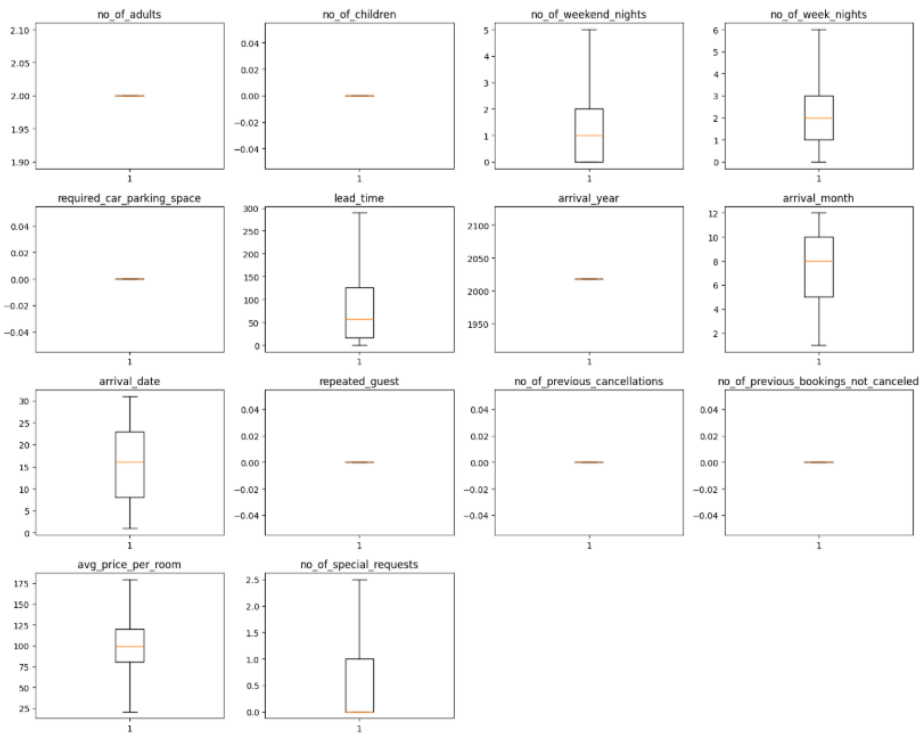
Data Preprocessing

Outlier Detection and Treatment

There are some Outliers in the dataset



The Outliers has been treated



Train - Test Split

```
Shape of Training set :  (25392, 24)
Shape of test set :  (10883, 24)

Percentage of classes in training set:
booking_status
0    0.673992
1    0.326008
Name: proportion, dtype: float64

Percentage of classes in test set:
booking_status
0    0.668566
1    0.331434
Name: proportion, dtype: float64
```

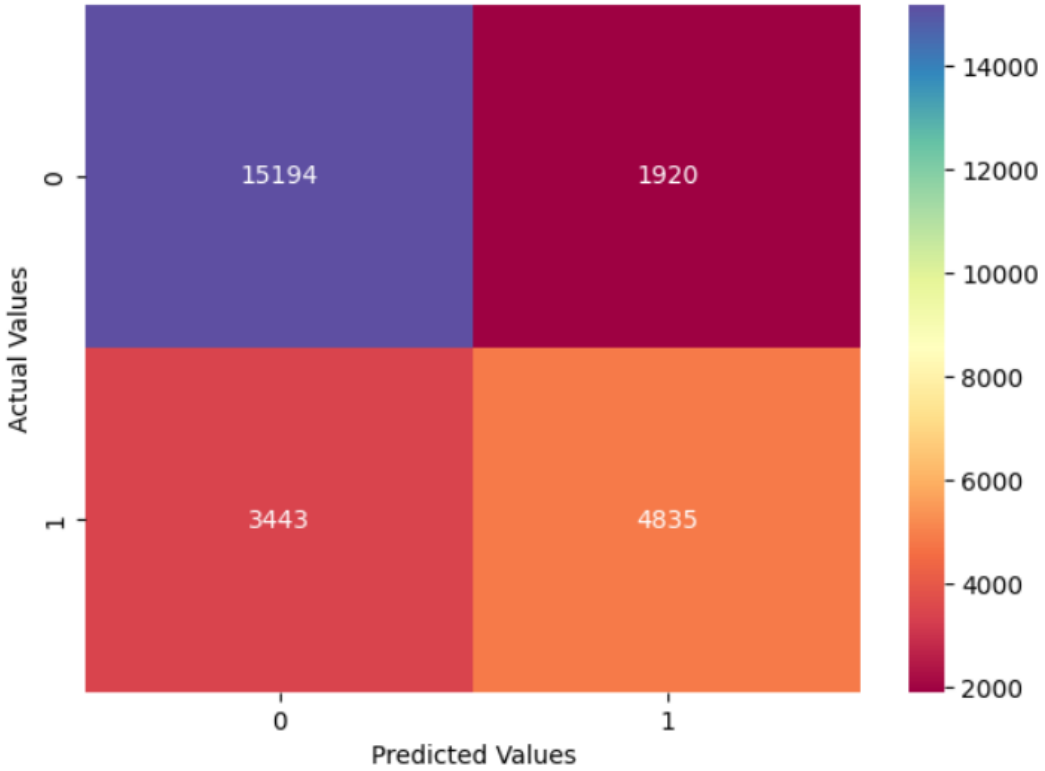
Model Building

Logistic Regression (stats-models)

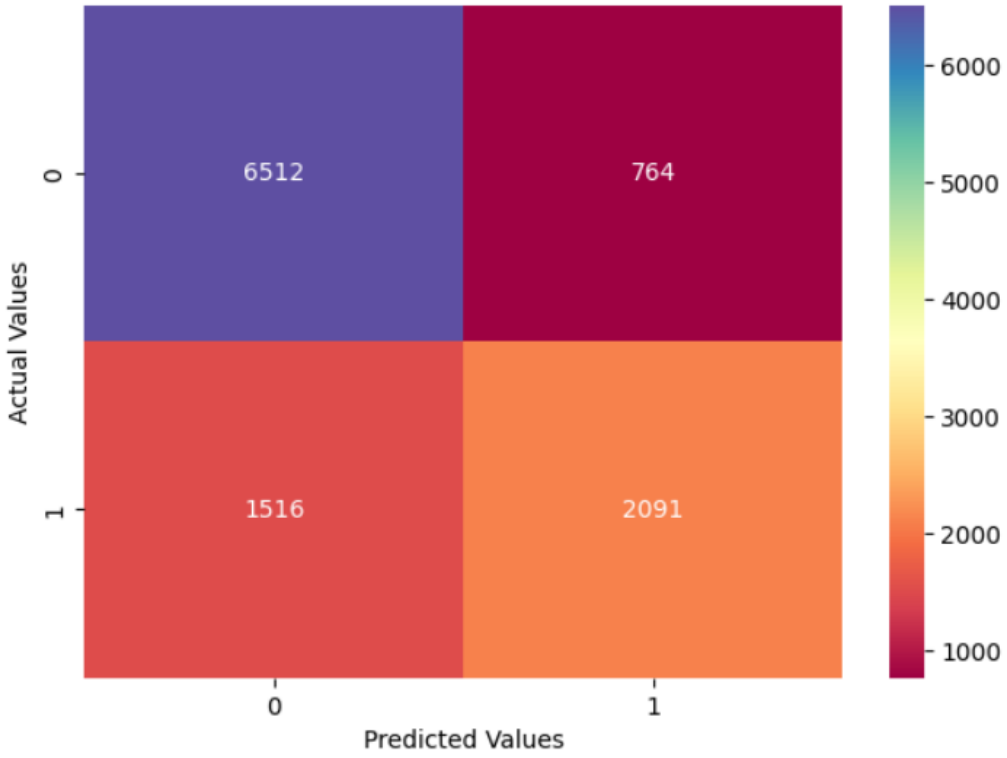
Logistic Regression (Default 0.5):

```
                           Logit Regression Results
==============================================================================
Dep. Variable:         booking_status   No. Observations:                25392
Model:                          Logit   Df Residuals:                    25370
Method:                           MLE   Df Model:                           21
Date:                Sun, 27 Apr 2025   Pseudo R-squ.:                  0.2955
Time:                        08:31:16   Log-Likelihood:                -11293.
converged:                       False   LL-Null:                       -16030.
Covariance Type:            nonrobust   LLR p-value:                     0.000
==================================================================================================
                                     coef    std err          z      P>|z|      [0.025      0.975]
--------------------------------------------------------------------------------------------------
const                             -1.4773    534.208     -0.003      0.998   -1048.506    1045.552
lead_time                          1.3099      0.021     63.116      0.000       1.269       1.351
arrival_month                     -0.1318      0.017     -7.570      0.000      -0.166      -0.098
arrival_date                       0.0274      0.017      1.658      0.097      -0.005       0.060
total_nights                       0.2223      0.031      7.271      0.000       0.162       0.282
average_price_per_night            0.3198      0.026     12.082      0.000       0.268       0.372
special_request_count             -1.0698      0.022    -48.812      0.000      -1.113      -1.027
is_long_stay                       0.0210      0.019      1.122      0.262      -0.016       0.058
weekend_stay                       0.0566      0.019      2.978      0.003       0.019       0.094
type_of_meal_plan_Meal Plan 2      0.1406      0.017      8.156      0.000       0.107       0.174
type_of_meal_plan_Meal Plan 3      0.1991    313.817      0.001      0.999    -614.871     615.269
type_of_meal_plan_Not Selected     0.0283      0.018      1.596      0.110      -0.006       0.063
room_type_reserved_Room_Type 2    -0.0788      0.017     -4.617      0.000      -0.112      -0.045
room_type_reserved_Room_Type 3     0.0175      0.030      0.591      0.555      -0.041       0.076
room_type_reserved_Room_Type 4     0.0666      0.018      3.729      0.000       0.032       0.102
room_type_reserved_Room_Type 5    -0.0027      0.017     -0.155      0.877      -0.037       0.031
room_type_reserved_Room_Type 6     0.0760      0.016      4.611      0.000       0.044       0.108
room_type_reserved_Room_Type 7     0.0555      0.018      3.086      0.002       0.020       0.091
market_segment_type_Complementary -4.2267   5152.639     -0.001      0.999   -1.01e+04    1.01e+04
market_segment_type_Corporate     -0.2203      0.062     -3.581      0.000      -0.341      -0.100
market_segment_type_Offline       -0.7036      0.116     -6.050      0.000      -0.932      -0.476
market_segment_type_Online         0.2035      0.121      1.678      0.093      -0.034       0.441
==================================================================================================
```

Confusion Matrix (Train):



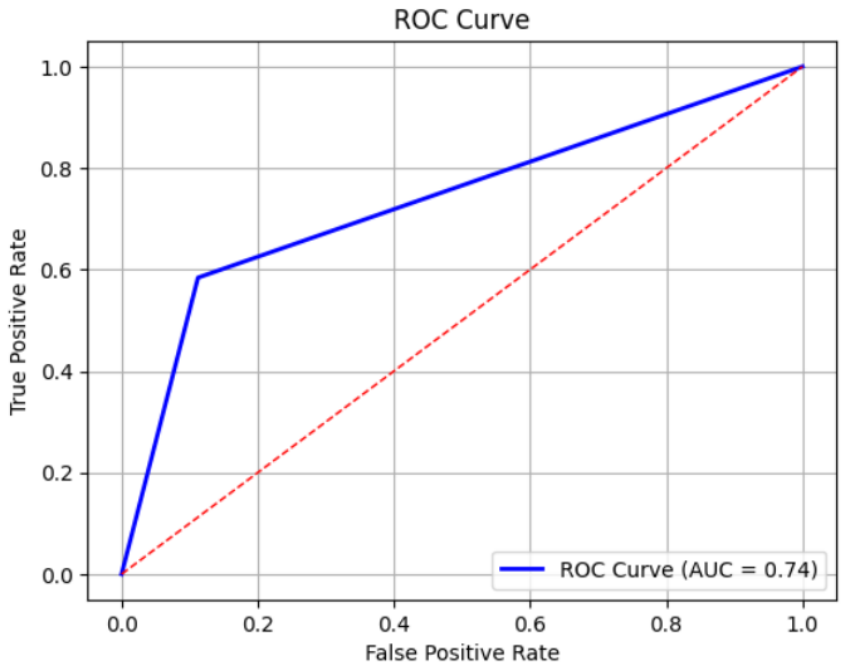
Confusion Matrix (Test):



- True Positives (TP): A person has Cancelled and the model predicted person has Cancelled the booking.
- True Negatives (TN): A person doesn't have Cancelled and the model predicted person doesn't have Cancelled the booking.
- False Positives (FP): The model predicted person has Cancelled but the person doesn't have Cancelled the booking.
- False Negatives (FN): The model predicted person doesn't have Cancelled but the person has Cancelled the booking.

Performance Metrics (Train):

	Accuracy	Recall	Precision	F1
0	0.788792	0.584078	0.715766	0.643252

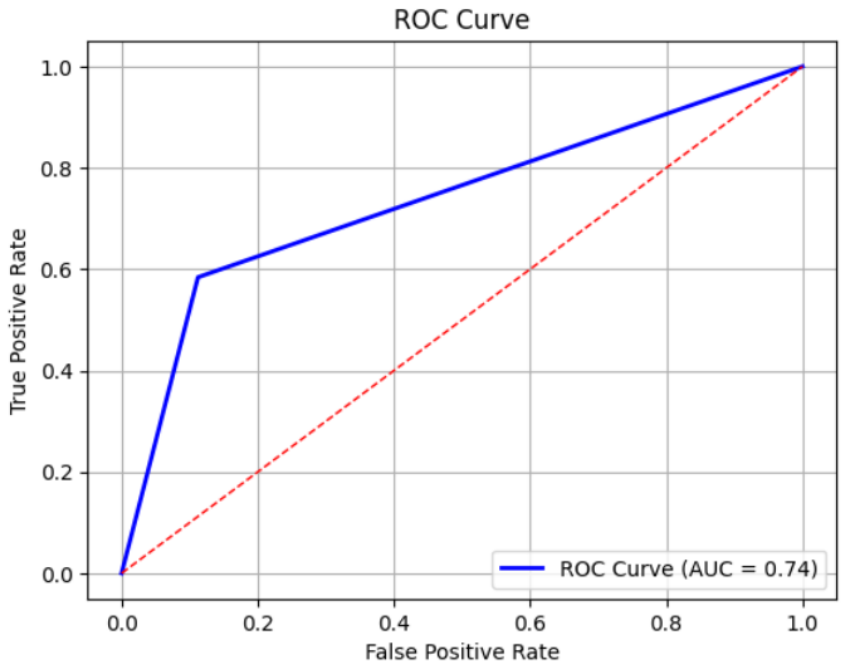


Insights:

- Shows decent performance.
- Recall = 58.40%
- It is not perfect, but not overfitting.

Performance Metrics (Test):

	Accuracy	Recall	Precision	F1
0	0.790499	0.579706	0.732399	0.647168

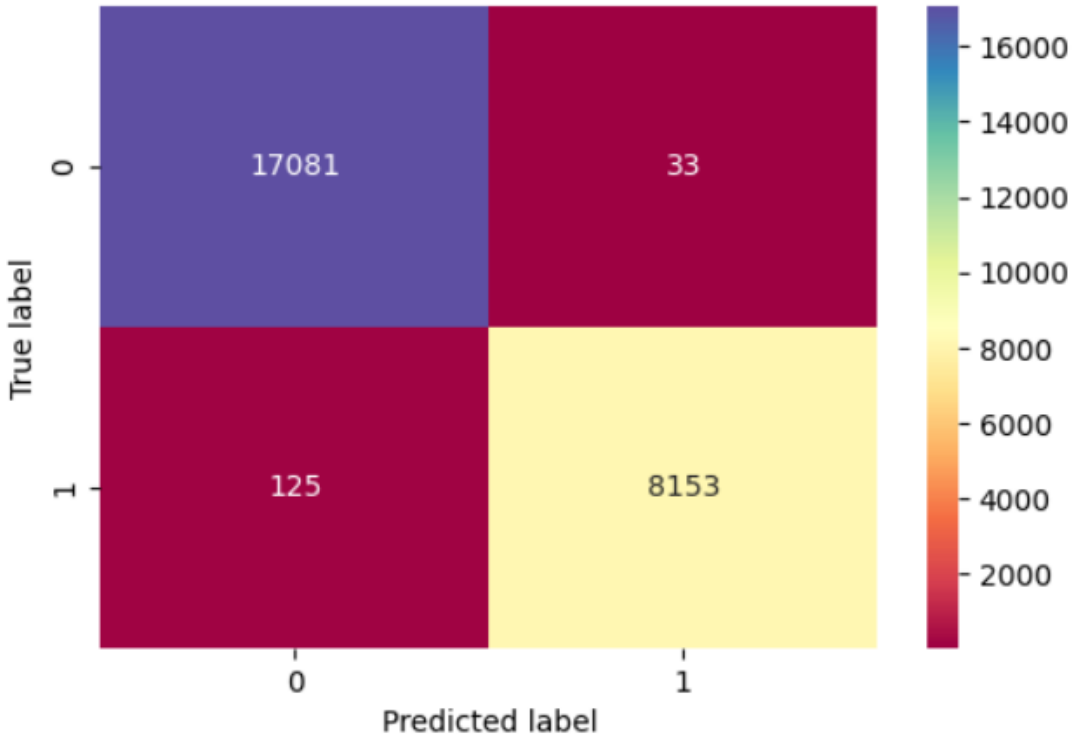


Insights:

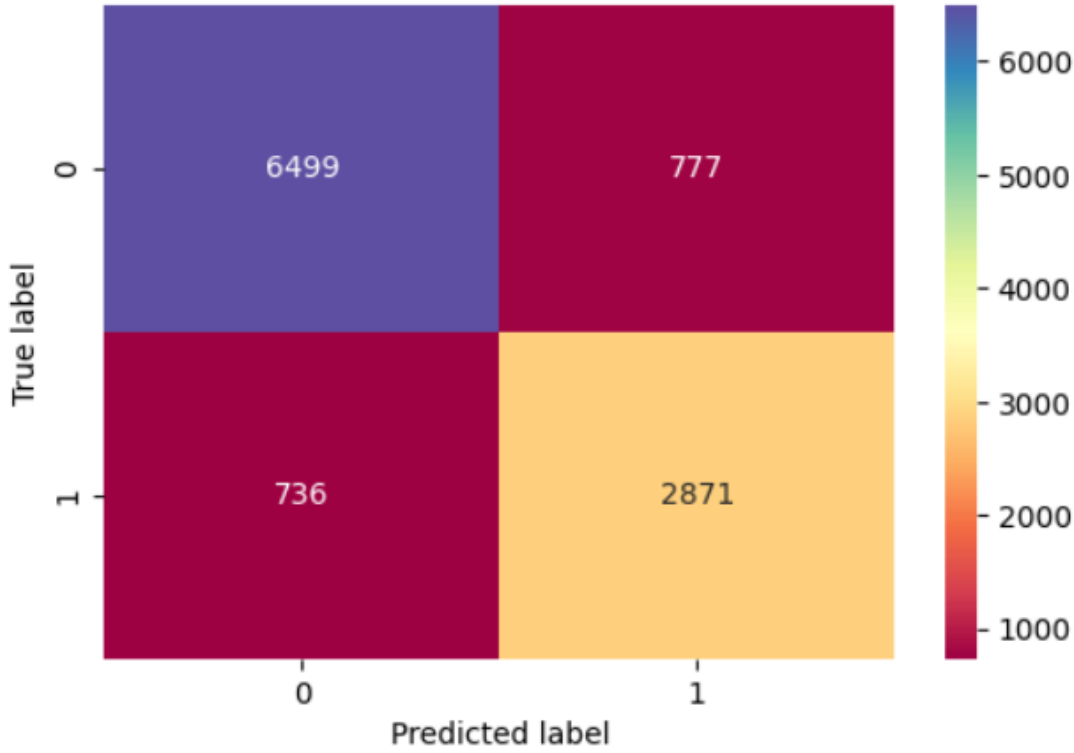
- Performance is stable from training to test.
- No sign of overfitting.
- It needs preprocessing or balancing.

Decision Tree Classifier (Sklearn)

Confusion Matrix (Train):



Confusion Matrix (Test):



Performance Metrics (Train):

	Accuracy	Recall	Precision	F1
0	0.993778	0.9849	0.995969	0.990403

Insights:

- Near-perfect scores.
- Classic sign of overfitting.

Performance Metrics (Test):

	Accuracy	Recall	Precision	F1
0	0.860976	0.795952	0.787007	0.791454

Insights:

The drop in metrics from training to test is huge and confirms overfitting, but it's still the better-performing model.

Final Observations:

- Logistic regression is not good without addressing imbalance or threshold tuning.
- Decision Tree requires regularization (pre-/post-pruning) to improve generalization.

Model Performance Improvement

Logistic Regression (stats-models)

Checking for Multicollinearity:

```
VIF values:

lead_time                              1.238603
arrival_month                          1.044660
arrival_date                           1.005782
total_nights                           3.462733
average_price_per_night                2.611296
special_request_count                  1.218679
is_long_stay                           1.423952
weekend_stay                           1.308326
type_of_meal_plan_Meal Plan 2          1.141377
type_of_meal_plan_Meal Plan 3          1.006281
type_of_meal_plan_Not Selected         1.214495
room_type_reserved_Room_Type 2         1.028952
room_type_reserved_Room_Type 3         1.004402
room_type_reserved_Room_Type 4         1.199634
room_type_reserved_Room_Type 5         1.024869
room_type_reserved_Room_Type 6         1.088547
room_type_reserved_Room_Type 7         1.039743
market_segment_type_Complementary      4.334972
market_segment_type_Corporate         17.012141
market_segment_type_Offline           63.581376
market_segment_type_Online            70.419815
dtype: float64
```

Variables with High Multicollinearity

- market_segment_type_Corporate: VIF = 17
- market_segment_type_Offline: VIF = 63.58

- market_segment_type_Online: VIF = 70.41

After removing Multicollinearity:

```
VIF values after removing high multicollinearity features:

const                                 1.000000
lead_time                             1.172380
arrival_month                         1.040715
arrival_date                          1.005581
total_nights                          3.312924
average_price_per_night               2.497920
special_request_count                 1.078661
is_long_stay                          1.408888
weekend_stay                          1.304627
type_of_meal_plan_Meal Plan 2         1.086953
type_of_meal_plan_Meal Plan 3         1.006141
type_of_meal_plan_Not Selected        1.093550
room_type_reserved_Room_Type 2        1.016156
room_type_reserved_Room_Type 3        1.004223
room_type_reserved_Room_Type 4        1.150332
room_type_reserved_Room_Type 5        1.016356
room_type_reserved_Room_Type 6        1.078364
room_type_reserved_Room_Type 7        1.039292
market_segment_type_Complementary     1.136131
dtype: float64
```

Checking insignificant variables (p-value > 0.05):

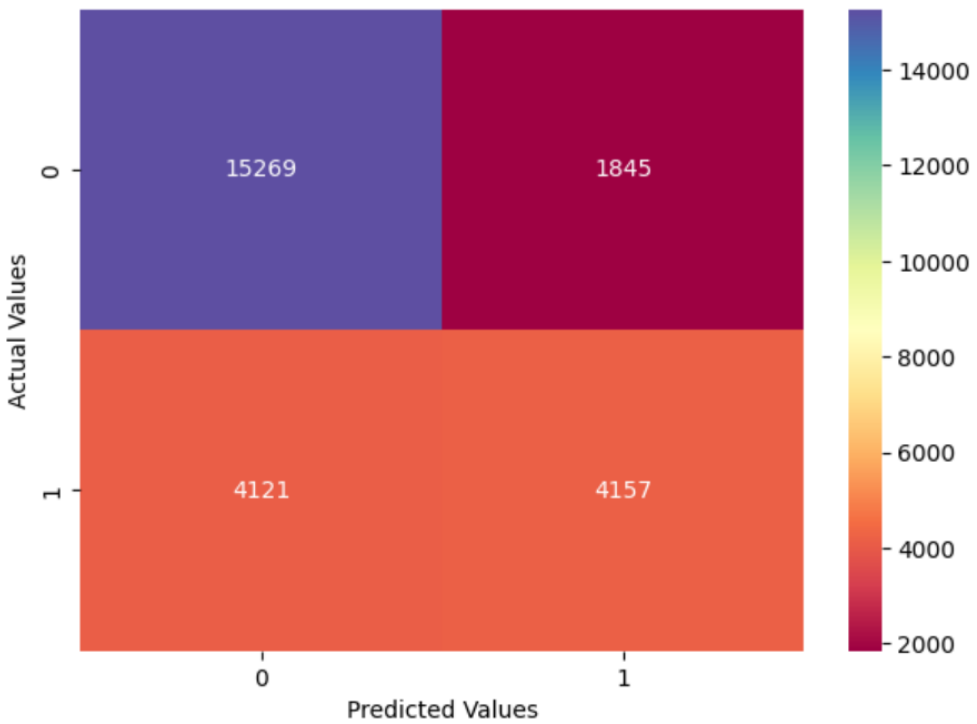
```
                           Logit Regression Results
==============================================================================
Dep. Variable:         booking_status   No. Observations:                25392
Model:                          Logit   Df Residuals:                    25370
Method:                           MLE   Df Model:                           21
Date:                Sat, 26 Apr 2025   Pseudo R-squ.:                  0.2955
Time:                        18:25:26   Log-Likelihood:                -11293.
converged:                       False   LL-Null:                       -16030.
Covariance Type:            nonrobust   LLR p-value:                     0.000
=================================================================================================
                                    coef    std err          z      P>|z|      [0.025      0.975]
-------------------------------------------------------------------------------------------------
const                            -1.4773    534.208     -0.003      0.998   -1048.506    1045.552
lead_time                         1.3099      0.021     63.116      0.000       1.269       1.351
arrival_month                    -0.1318      0.017     -7.570      0.000      -0.166      -0.098
arrival_date                      0.0274      0.017      1.658      0.097      -0.005       0.060
total_nights                      0.2223      0.031      7.271      0.000       0.162       0.282
average_price_per_night           0.3198      0.026     12.082      0.000       0.268       0.372
special_request_count            -1.0698      0.022    -48.812      0.000      -1.113      -1.027
is_long_stay                      0.0210      0.019      1.122      0.262      -0.016       0.058
weekend_stay                      0.0566      0.019      2.978      0.003       0.019       0.094
type_of_meal_plan_Meal Plan 2     0.1406      0.017      8.156      0.000       0.107       0.174
type_of_meal_plan_Meal Plan 3     0.1991    313.817      0.001      0.999    -614.871     615.269
type_of_meal_plan_Not Selected    0.0283      0.018      1.596      0.110      -0.006       0.063
room_type_reserved_Room_Type 2   -0.0788      0.017     -4.617      0.000      -0.112      -0.045
room_type_reserved_Room_Type 3    0.0175      0.030      0.591      0.555      -0.041       0.076
room_type_reserved_Room_Type 4    0.0666      0.018      3.729      0.000       0.032       0.102
room_type_reserved_Room_Type 5   -0.0027      0.017     -0.155      0.877      -0.037       0.031
room_type_reserved_Room_Type 6    0.0760      0.016      4.611      0.000       0.044       0.108
room_type_reserved_Room_Type 7    0.0555      0.018      3.086      0.002       0.020       0.091
market_segment_type_Complementary -4.2267   5152.639     -0.001      0.999   -1.01e+04    1.01e+04
market_segment_type_Corporate    -0.2203      0.062     -3.581      0.000      -0.341      -0.100
market_segment_type_Offline      -0.7036      0.116     -6.050      0.000      -0.932      -0.476
market_segment_type_Online        0.2035      0.121      1.678      0.093      -0.034       0.441
=================================================================================================
```

Removing insignificant variables (p-value > 0.05):

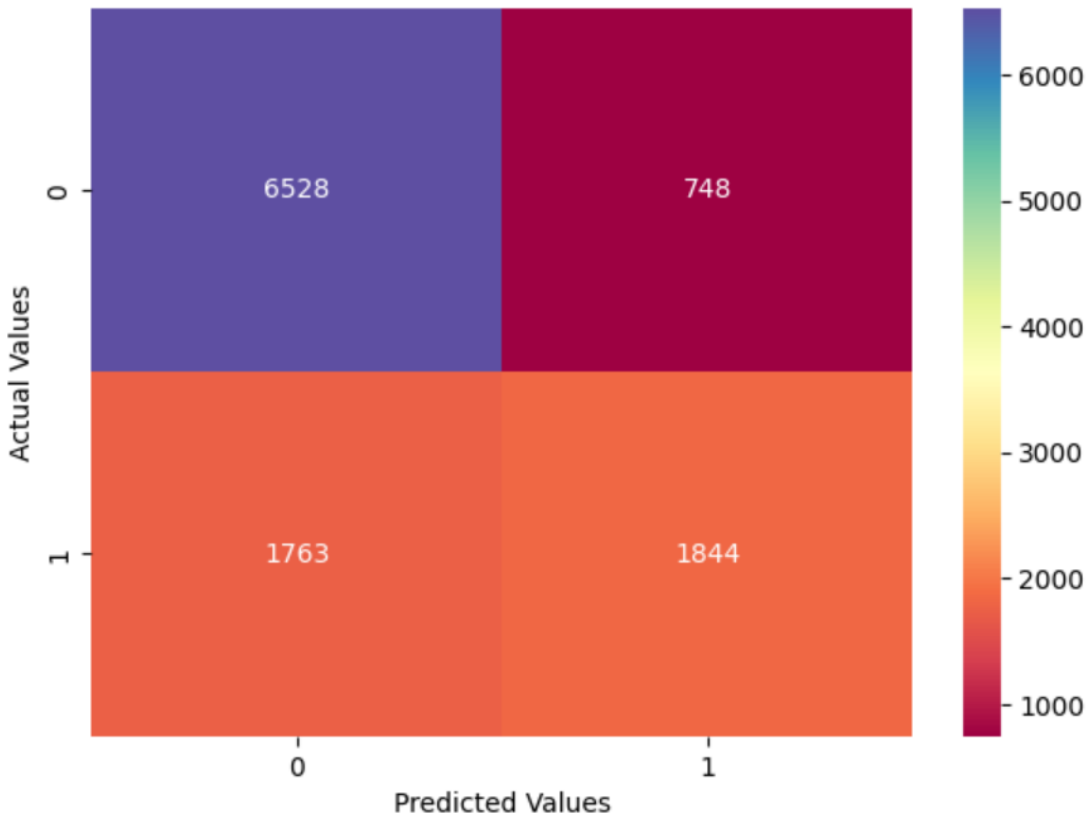
```
                           Logit Regression Results
==============================================================================
Dep. Variable:         booking_status   No. Observations:                25392
Model:                          Logit   Df Residuals:                    25380
Method:                           MLE   Df Model:                           11
Date:                Sat, 26 Apr 2025   Pseudo R-squ.:                  0.2362
Time:                        18:54:25   Log-Likelihood:                -12245.
converged:                       True   LL-Null:                       -16030.
Covariance Type:            nonrobust   LLR p-value:                     0.000
==================================================================================================
                                     coef    std err          z      P>|z|      [0.025      0.975]
--------------------------------------------------------------------------------------------------
const                             -0.9720      0.017    -57.304      0.000      -1.005      -0.939
lead_time                          1.1428      0.018     62.907      0.000       1.107       1.178
arrival_month                     -0.1715      0.017    -10.285      0.000      -0.204      -0.139
arrival_date                       0.0301      0.016      1.908      0.056      -0.001       0.061
total_nights                       0.4067      0.024     17.118      0.000       0.360       0.453
average_price_per_night            0.5019      0.023     21.559      0.000       0.456       0.548
special_request_count             -0.7687      0.019    -39.827      0.000      -0.806      -0.731
weekend_stay                       0.1003      0.018      5.532      0.000       0.065       0.136
type_of_meal_plan_Not Selected     0.2567      0.016     15.905      0.000       0.225       0.288
room_type_reserved_Room_Type 4     0.2271      0.016     13.895      0.000       0.195       0.259
room_type_reserved_Room_Type 6     0.1441      0.015      9.339      0.000       0.114       0.174
room_type_reserved_Room_Type 7     0.0654      0.017      3.963      0.000       0.033       0.098
==================================================================================================
```

Performance Metrics of the final model

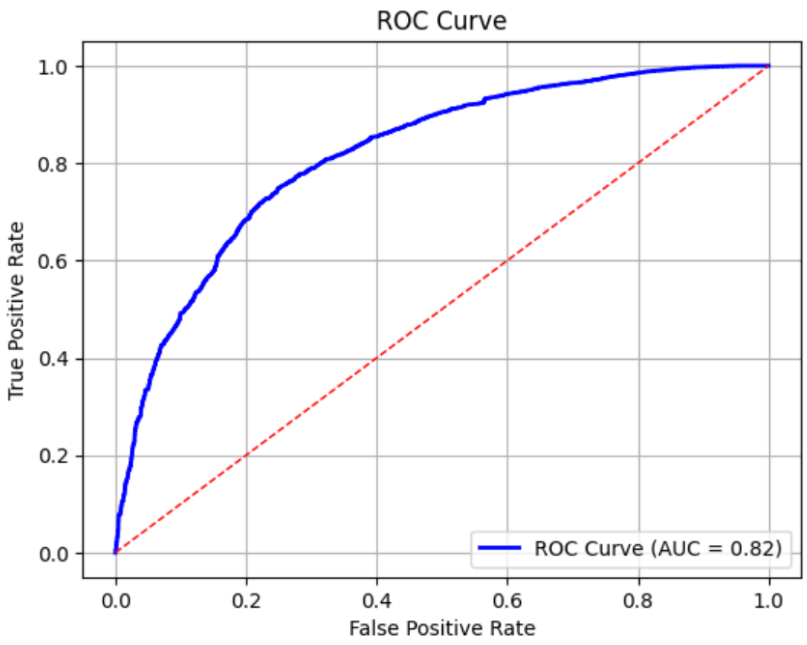
Confusion Matrix (Train):



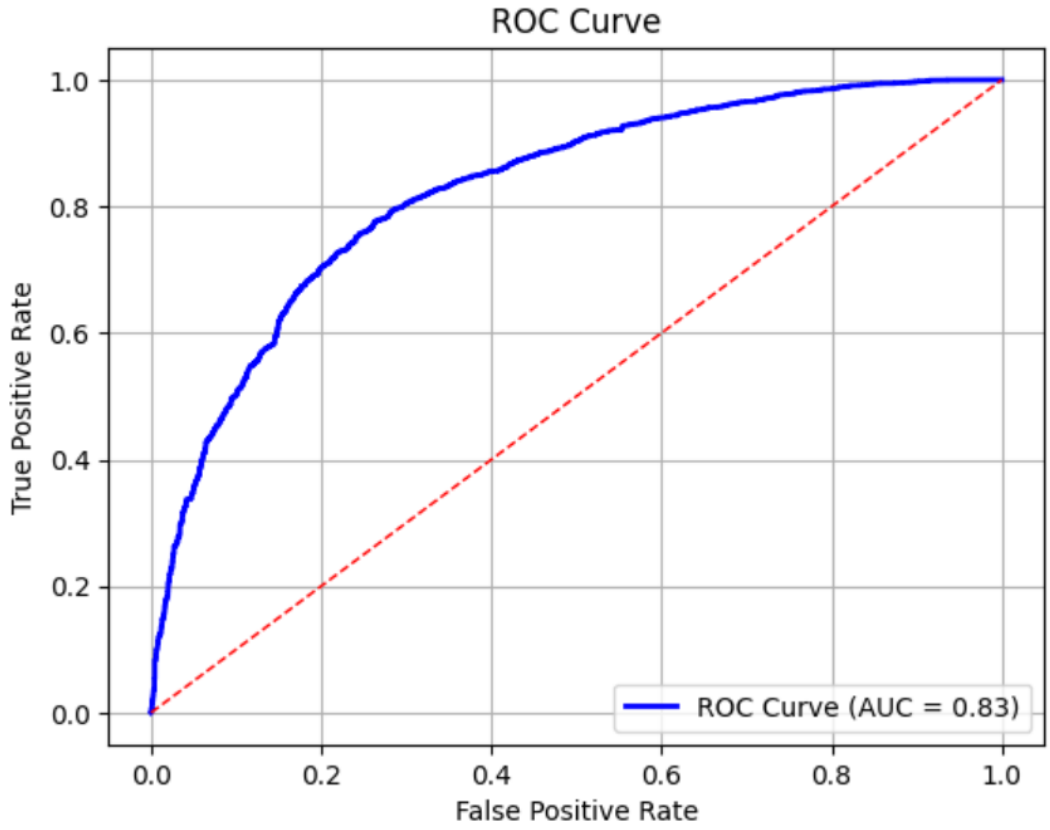
Confusion Matrix (Test):



ROC – AUC Curve (Train):



ROC – AUC Curve (Test):



Optimal Threshold (0.29):

Performance Metrics (Train):

	Accuracy	Recall	Precision	F1
0	0.793321	0.533511	1.0	0.695803

Performance Metrics (Test):

	Accuracy	Recall	Precision	F1
0	0.834696	0.590298	1.0	0.742374

Training performance comparison:

```
Training performance comparison:
```

	Logistic Model Default (Stats)	Logistic Model Threshold of 0.29
Accuracy	0.788792	0.793321
Recall	0.584078	0.533511
Precision	0.715766	1.000000
F1	0.643252	0.695803

Insights (Training)

Logistic Model Default (Threshold = 0.5):

- Shows decent baseline performance.
- Recall = 58.40%
- Precision = 72%
- F1 = 64.32% balance between recall and precision.

Not overfitting, as performance is reasonable without extreme values.

Logistic Model with Threshold (0.29):

- Improved performance across all metrics.
- Accuracy = 79.33% significantly better than default.
- Recall = 53.3%
- Precision = 100% (very high confidence).
- F1 = 69.58% strong overall performance.

It has made good progress.

Test performance comparison:

	Logistic Model Default (Stats)	Logistic Model Threshold of 0.29
Accuracy	0.788792	0.834696
Recall	0.584078	0.590298
Precision	0.715766	1.000000
F1	0.643252	0.742374

Insights (Test):

Logistic Model Default (Threshold = 0.5):

- Decent generalization from training to test set.
- Accuracy = 78.87%
- Recall = 58.40%
- Precision = 72%
- F1 = 64.32% balance performance.

No signs of overfitting.

Logistic Model with Threshold (0.29):

- Better overall performance on test data.
- Accuracy = 83.46% significantly better than default.
- Recall = 59.02%
- Precision = 100% (very high confidence).
- F1 = 74.23% strong overall performance.

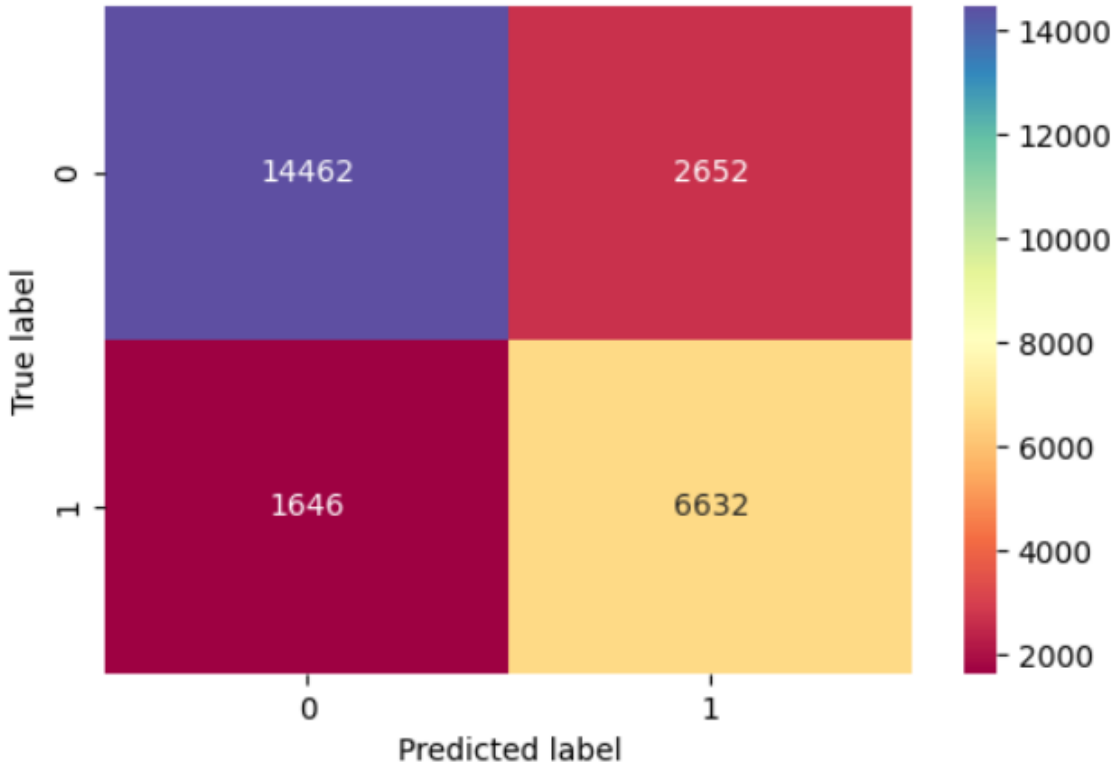
It has made good progress than training set.

Decision Tree Pre-pruning (Sklearn)

Model Performance

```
DecisionTreeClassifier
DecisionTreeClassifier(class_weight='balanced', max_depth=12, max_leaf_nodes=50,
                       random_state=42)
```

Confusion Matrix (Train):



Performance metric on train:

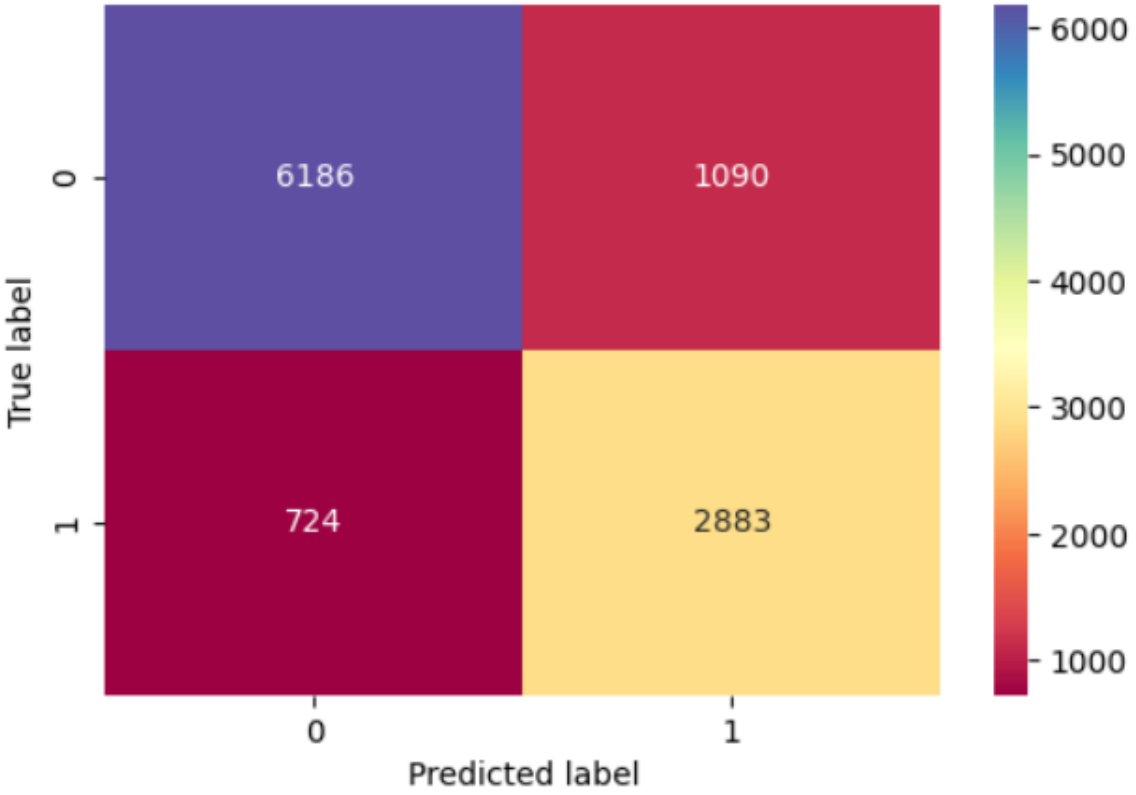
	Accuracy	Recall	Precision	F1
0	0.830734	0.80116	0.714347	0.755267

Insights (Train):

- Accuracy = 83.07%
- Recall = 80.11%
- Precision = 71.14%
- F1 = 76% balanced but moderate performance.

Pre-pruning helps to control complexity and avoid overfitting.

Confusion Matrix (Test):



Performance metric on test:

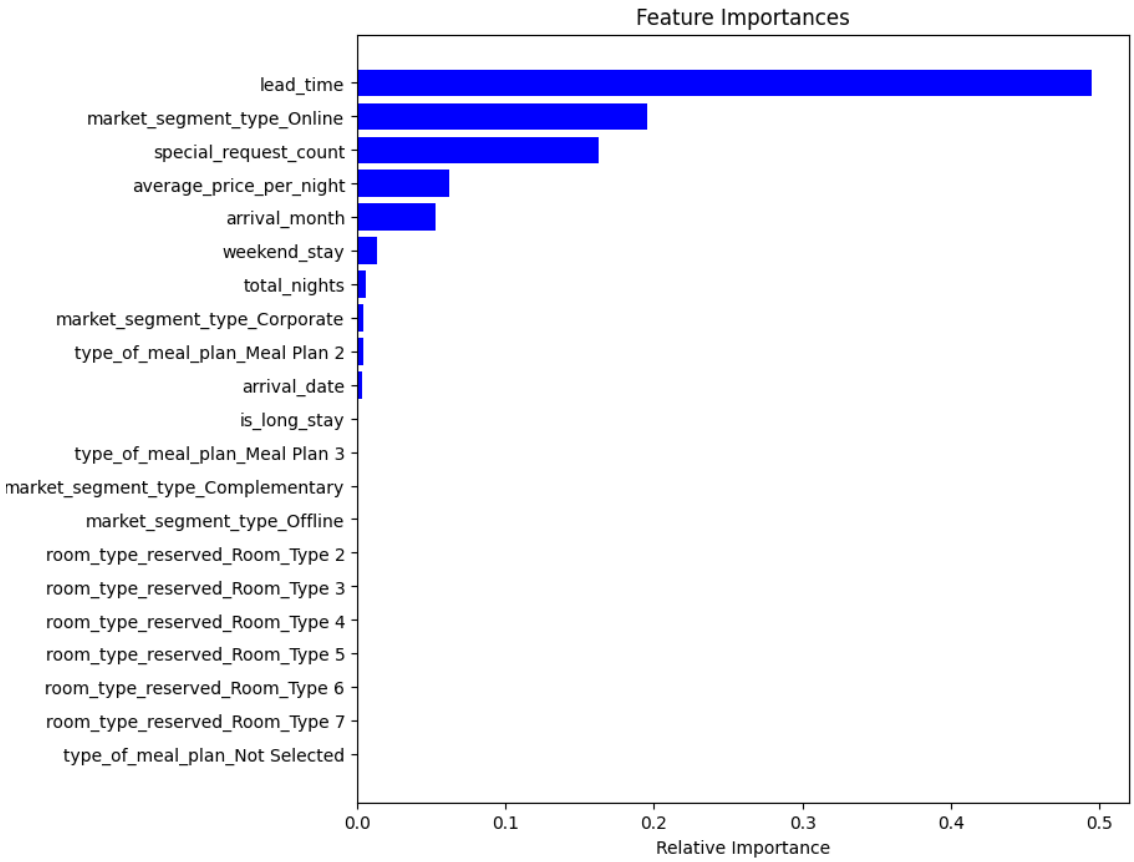
	Accuracy	Recall	Precision	F1
0	0.833318	0.799279	0.725648	0.760686

Insights (Test):

- Accuracy = 83.33% balanced.
- Recall = 79.27%
- F1 = 76.06 moderate, with slightly better balance than post-pruning.
- Precision = 72.56

Reduced overfitting compared to the default tree.

Feature Importance:



1. **lead_time**

   Most important by far (almost **50%** relative importance). (Typically, bookings made far in advance have higher chances of cancellation.)

2. **market_segment_type_Online**
   Online bookings are the second biggest factor.
   Online customers might behave differently — more flexible to cancel compared to walk-ins or corporate bookings.

3. **special_request_count**
   Special requests (like extra beds, late checkouts) are important too.
   Customers who make special requests might be more serious about showing up, or very selective.

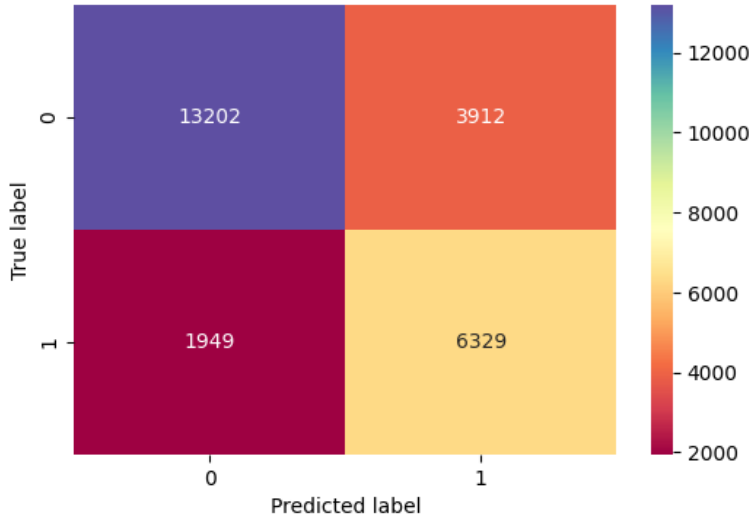
4. **average_price_per_night**
   Higher priced bookings impact cancellations.
   Possibly, expensive stays cause more cancellations if customers change their mind.

Decision Tree Post-pruning (Sklearn)

Model Performance

```
DecisionTreeClassifier
DecisionTreeClassifier(ccp_alpha=0.013157894736842105, random_state=42)
```

Confusion Matrix (Train):



Performance metric on train:

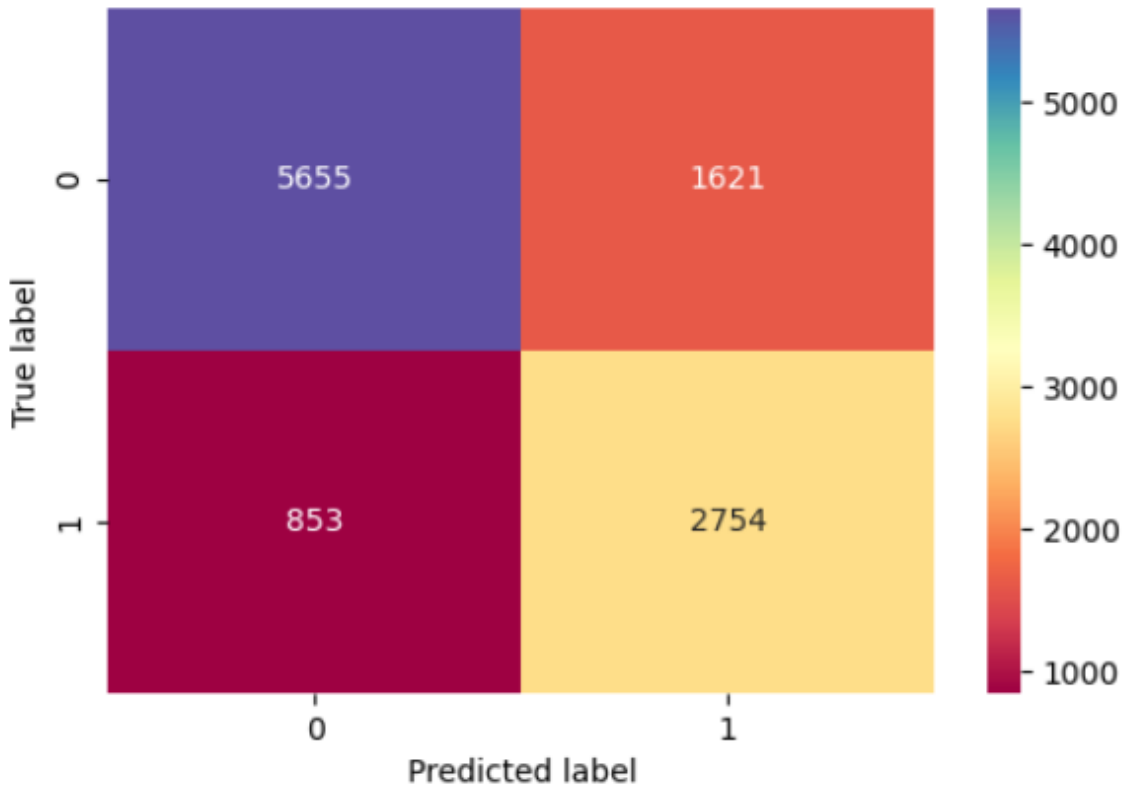
	Accuracy	Recall	Precision	F1
0	0.769179	0.764557	0.618006	0.683514

Insights (Train):

- Accuracy = 76.9%, similar to pre-pruning.
- Recall = 76.4%,
- Precision = 61.80
- F1 = 68.3% balanced but moderate performance.

Post-pruning also reduces overfitting, though it may slightly reduce recall.

Confusion Matrix (Test):



Performance metric on test:

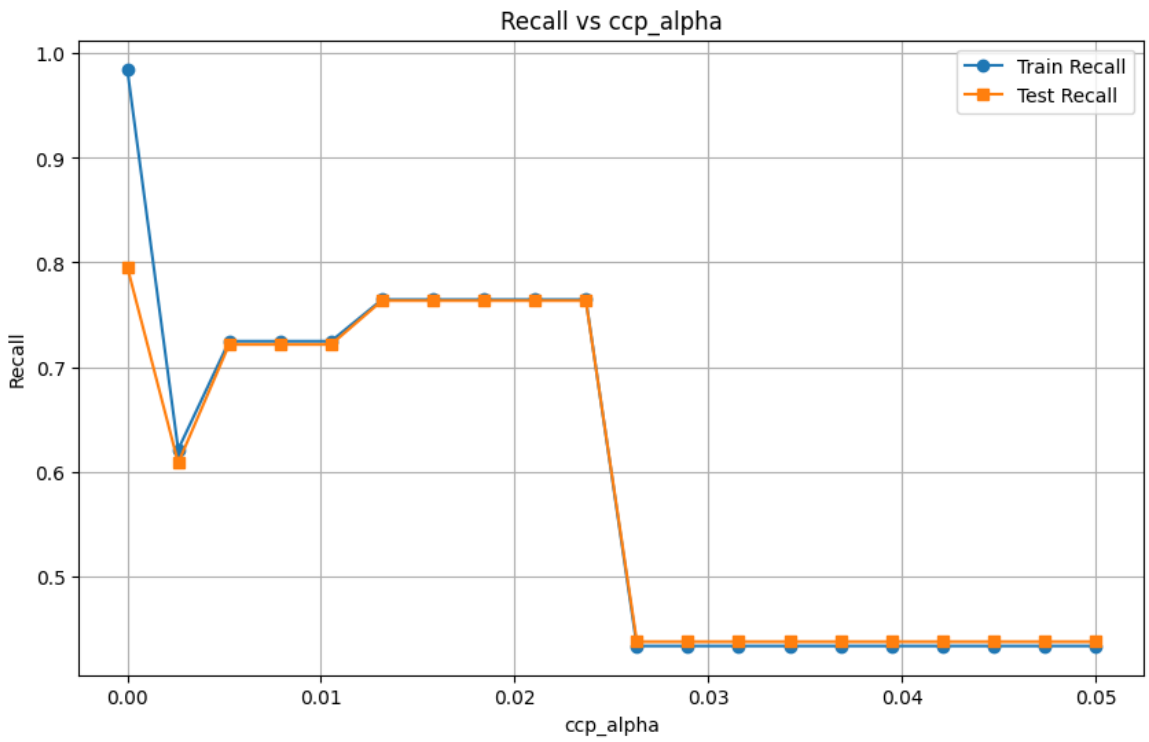
	Accuracy	Recall	Precision	F1
0	0.772673	0.763515	0.629486	0.690053

Insights (Test):

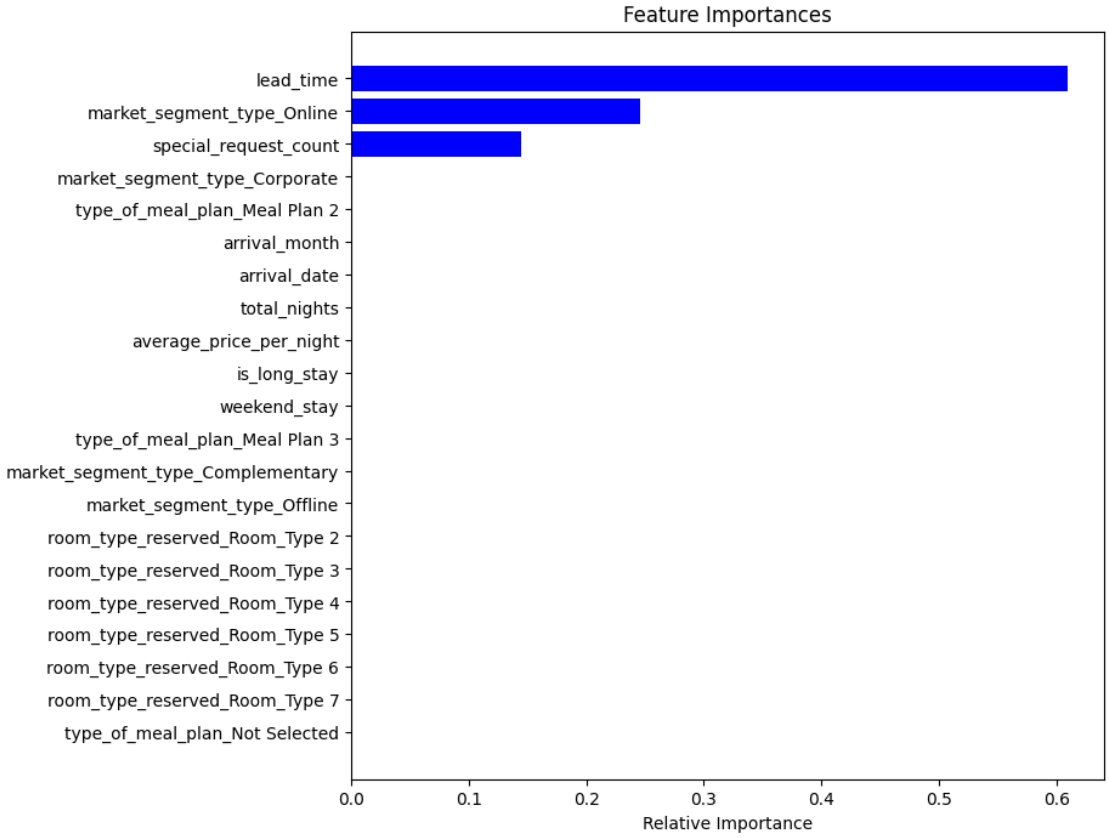
- Accuracy = 77.26%, comparable to pre-pruning.
- Recall = 76.35%, Precision = 63%
- F1 = 69%, just below pre-pruning.

Helps keep the model simpler, at the slight cost of performance.

Recall vs CCP alpha:



Feature Importance:



1) **lead_time**

   Most important by far (almost **50%** relative importance). (Typically, bookings made far in advance have higher chances of cancellation.)

2) **market_segment_type_Online**

   Online bookings are the second biggest factor.
   Online customers might behave differently — more flexible to cancel compared to walk-ins or corporate bookings.

3) **special_request_count**

   Special requests (like extra beds, late checkouts) are important

too.
Customers who make special requests might be more serious about showing up, or very selective.

Model Performance Comparison and Final Model Selection

Training Comparison

Training performance comparison:

	Logistic Model Default (Stats)	Logistic Model Threshold of 0.29	Decision Tree Default (Sklearn)	Decision Tree Pre-pruning	Decision Tree Post-pruning
Accuracy	0.788792	0.793321	0.993778	0.830734	0.769179
Recall	0.584078	0.533511	0.984900	0.801160	0.764557
Precision	0.715766	1.000000	0.995969	0.714347	0.618006
F1	0.643252	0.695803	0.990403	0.755267	0.683514

Test Comparison

Test performance comparison:

	Logistic Model Default (Stats)	Logistic Model Threshold of 0.29	Decision Tree Default (Sklearn)	Decision Tree Pre-pruning	Decision Tree Post-pruning
Accuracy	0.790499	0.834696	0.860976	0.833318	0.772673
Recall	0.579706	0.590298	0.795952	0.799279	0.763515
Precision	0.732399	1.000000	0.787007	0.725648	0.629486
F1	0.647168	0.742374	0.791454	0.760686	0.690053

Insights

Logistic Regression (Default Threshold 0.5)

Training Performance: Moderate performance across all metrics.
Test Performance: Accuracy and precision are good, but recall is slightly lower.
Observation: Balanced but not the best performer. Slight

underfitting.

Logistic Regression (Threshold = 0.29)

Training Performance: Improvement in all metrics after adjusting threshold.
Test Performance: - Highest accuracy (83%), Recall (59%) and F1 score (74%). - Perfect precision (1.0), suggesting zero false positives.
Observation: A well-defined threshold significantly improves performance while avoiding overfitting.

Decision Tree (Default - Sklearn)

Training Performance: Extremely high across the board. Likely overfitting.
Test Performance: - High recall (86%) and good F1 score (79%). - Accuracy is decent but lower than logistic with threshold.
Observation: Performs very well but tends to overfit, evident from the large drop from training to test scores.

Decision Tree (Pre-pruning)

Training Performance: Regularized, more balanced.
Test Performance: Metrics are improved compared to post-pruning.
Observation: Recall still very strong (80%), slightly simpler tree.

Decision Tree (Post-pruning)

Training Performance: Consistently lower recall and F1 scores.
Test Performance: Consistently lower recall and F1 scores.
Observation: Excessive regularization may have reduced model capacity too much.

Model Recommendation

Decision Tree (Pre-pruning)

- **High Recall (80%)** - Catches cancellations better than Logistic Regression (important for business).
- **Good Precision (73%)** - Reasonably few false alarms.
- **F1 Score is Higher (76%)** - Balanced metric (better than Logistic Model at threshold 0.31).
- **Generalized Model** - Pre-pruning prevents overfitting compared to default decision tree.

Overall, the **Decision Tree Pre-pruning** is **more balanced**, **robust**, and **practical** for deployment.

Actionable Insights & Recommendations

Actionable Insights:

- Lead Time is the most influential factor affecting customer behavior.
- Market Segment Type (Online) bookings play a major role in customer outcomes.
- Special Request Count has a strong impact on guest satisfaction and booking retention.
- Average Price Per Night affects customer decision-making significantly.
- Arrival Month influences booking patterns and customer behavior.
- Weekend Stay and Long Stay factors have minor, but notable, effects.

- Corporate Market Segment shows measurable but smaller influence compared to online bookings.

Recommendations:

1. **Optimize Lead Time:**
    - For moderate lead times, offer early discounts.
    - Introduce flexible cancellation policies for very early bookings.
2. **Enhance Online Booking Experience:**
    - Improve website/app UX/UI.
3. **Personalize Services Based on Special Requests:**
    - Prioritize fulfilling special requests to improve guest experience.
    - Offer customizable add-on packages during the booking process.
4. **Implement Smart Pricing Strategies:**
    - Bundle services (e.g., breakfast, spa) instead of just offering discounts.
5. **Leverage Seasonality (Arrival Month):**
    - Design seasonal offers for low-demand months.
    - Adjust staffing and inventory in advance based on forecasted demand.
6. **Encourage Weekend and Long Stays:**
    - Offer weekend getaway packages.
    - Provide discounted rates for longer stays, especially during off-peak periods.